

# [ME315A] Lab07

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## SQLite

O **SQLite** é uma biblioteca embutida em um processo único que é autônomo, não necessita de servidor e exige zero de configuração. O código é de domínio público e seu uso é gratuito em qualquer tipo de utilização. Trata-se de um sistema SQL completo, capaz de utilizar múltiplas tabelas, índices, gatilhos e visões. Ele funciona, virtualmente, em qualquer plataforma (incluindo móveis) e pesa cerca de 600KB.

## Objetivos

Ao fim deste laboratório, você deverá ser capaz de:

- Conectar-se a um banco de dados do tipo SQLite utilizando o R como interface;
- Explorar quais são as tabelas disponíveis no referido banco de dados;
- Identificar quais são as colunas de uma dada tabela existente no banco de dados;
- Realizar pesquisas simples;
- Extrair registros do banco de dados e armazená-las em objetos do R;
- Realizar algumas pesquisas complexas, utilizando `WHERE`, `INNER JOIN`.
- Criar novas tabelas no banco de dados.

## Fonte do Problema

Estes dados são de uma iniciativa de publicidade de desempenho de alunos e professores nas Universidades Americanas. A UW-Madison disponibilizou os seus dados online (no Office of the Registrar: <https://registrar.wisc.edu/>), e alguém trouxe os dados para uma competição no Kaggle (<https://www.kaggle.com/Madgrades/uw-madison-courses>). Nesta atividades, examinaremos disciplinas oferecidas no assunto de Estatística pela referida Universidade.

## Observações

- Esta tarefa não deve ser realizada no site `jupyter.ime.unicamp.br` ;
- Você deve realizar o download dos dados para o computador que estiver utilizando e, então, iniciar a atividade;
- Recomenda-se a utilização dos seguintes pacotes:
  - `RSQLite`
- Toda a atividade deve ser realizada utilizando-se apenas de `SQLite` (i.e., não utilize `dbplyr`).

## Atividade

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(RSQLite)

caminho <- '/cloud/project'

caminho_completo <- file.path(caminho, 'database.sqlite3')

conn <- dbConnect(SQLite(), caminho_completo)
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
professores <- "
  SELECT DISTINCT instructors.name AS PROFESSOR FROM subjects
  INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
  INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid
  INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
  INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid
  INNER JOIN instructors
    ON teachings.instructor_id = instructors.id
  WHERE subjects.abbreviation = 'STAT'
  ORDER BY instructors.name
"
```

```
dbGetQuery(conn, professores)
```

|    | PROFESSOR                 |
|----|---------------------------|
| 1  | ABIGAIL BENZINE           |
| 2  | ADIN-CRISTIAN ANDREI      |
| 3  | AKICHIKA OZEKI            |
| 4  | ALAN HUANG                |
| 5  | ALBERTO DEL PIA           |
| 6  | ALBRECHT KLEMM            |
| 7  | ALEXANDER COVINGTON       |
| 8  | ALEXANDER FISH            |
| 9  | ALEXANDER KISELEV         |
| 10 | ALYSSA DIGILIO            |
| 11 | ANATOLE BECK              |
| 12 | ANDREJ ZLATOS             |
| 13 | ANDREW LESLIE             |
| 14 | ANNE BRUCKNER             |
| 15 | ANQI SHI                  |
| 16 | ANRU ZHANG                |
| 17 | ARNOLD MILLER             |
| 18 | AUGUST JENSEN             |
| 19 | BEHZAD AALIPUR            |
| 20 | BEHZAD AALIPUR HAFSHEJANI |
| 21 | BEN ADAM HAALAND          |
| 22 | BENEDEK VALKO             |
| 23 | BENJAMIN RECHT            |
| 24 | BI CHENG WU               |
| 25 | BIN DAI                   |
| 26 | BIN ZHANG                 |
| 27 | BIN ZHU                   |
| 28 | BINGYING XIE              |
| 29 | BO HUANG                  |
| 30 | BO YANG                   |
| 31 | BOWEN HU                  |
| 32 | BRET HANLON               |
| 33 | BRET LARGET               |
| 34 | BRIAN YANDELL             |
| 35 | BROOK LUERS               |
| 36 | CECILE ANE                |
| 37 | CHAN PARK                 |
| 38 | CHANHAN HSU               |

|    |                          |
|----|--------------------------|
| 39 | CHAOQUN MEI              |
| 40 | CHAOYANG YU              |
| 41 | CHELSEY GREEN            |
| 42 | CHEN CHENG               |
| 43 | CHEN JING                |
| 44 | CHENGNING ZHANG          |
| 45 | CHENLIANG XU             |
| 46 | CHENSHENG KUANG          |
| 47 | CHENXI LI                |
| 48 | CHIA-CHIEH LIN           |
| 49 | CHIEN-WEI CHEN           |
| 50 | CHRISTINA M. KENDZIORSKI |
| 51 | CHRISTINE SORKNESS       |
| 52 | CHRISTOPHER WAGNER       |
| 53 | CLAIRE BOBST             |
| 54 | CLAUDIA SOLIS LEMUS      |
| 55 | COLE COOK                |
| 56 | COLIN LONGHURST          |
| 57 | CRYSTAL CHEN             |
| 58 | CUICUI QI                |
| 59 | CUIZE HAN                |
| 60 | DANIELE CAPPELLETTI      |
| 61 | DAVID ANDERSON           |
| 62 | DAVID DEMETS             |
| 63 | DAVID GRIFFEATH          |
| 64 | DEBRAJ DAS               |
| 65 | DEREK BEAN               |
| 66 | DEREK NORTON             |
| 67 | DEYUAN JIANG             |
| 68 | DIETRICH UHLENBROCK      |
| 69 | DONALD PORTER            |
| 70 | DONG XIA                 |
| 71 | DONGGYU KIM              |
| 72 | DONGHYUN LEE             |
| 73 | DOUGLAS M. BATES         |
| 74 | DUY NGUYEN               |
| 75 | DUZHE WANG               |
| 76 | EDWARD ERKER             |
| 77 | ELOISA D CHAVAS          |
| 78 | ERICA LEE DEADMAN        |
| 79 | ERIK NORDHEIM            |
| 80 | FAN GAO                  |
| 81 | FAN YANG                 |

|     |                               |
|-----|-------------------------------|
| 82  | FANG FANG                     |
| 83  | FANGFANG WANG                 |
| 84  | FLORIAN BERTRAND              |
| 85  | FREDERICK BOEHM               |
| 86  | GARVESH RASKUTTI              |
| 87  | GARY HOWARD SCHROEDER         |
| 88  | GENG LI                       |
| 89  | GINA BENNINGER                |
| 90  | GINA OH                       |
| 91  | GONZALO CONTADOR              |
| 92  | GRACE WAHBA                   |
| 93  | GREGORIO MORENO-FLORES        |
| 94  | GREGORY SHINAULT              |
| 95  | GUANHUA CHEN                  |
| 96  | GUANNAN SUN                   |
| 97  | GUILHERME ROSA                |
| 98  | GUILHERME VIEIRA NUNES LUDWIG |
| 99  | GUN WOONG PARK                |
| 100 | HAN CHEN                      |
| 101 | HAO CHEN                      |
| 102 | HAO TENG                      |
| 103 | HAO ZHENG                     |
| 104 | HAO ZHOU                      |
| 105 | HAODA FU                      |
| 106 | HAOYANG FAN                   |
| 107 | HEATHER MARIE BRAZEAU         |
| 108 | HUAIBAO FENG                  |
| 109 | HUI WANG                      |
| 110 | HUIKUN ZHANG                  |
| 111 | HUILIN HU                     |
| 112 | HYEBIN SONG                   |
| 113 | HYUNSEUNG KANG                |
| 114 | IAN BRANSTAD RILEY            |
| 115 | ISMOR FISCHER                 |
| 116 | JAMES ANDERSON                |
| 117 | JAMES D KUELBS                |
| 118 | JARED BROWN                   |
| 119 | JASON P FINE                  |
| 120 | JASON RICHARD SWANSON         |
| 121 | JEE YEON KIM                  |
| 122 | JEEA CHOI                     |
| 123 | JENNIFER BIRSTLER             |
| 124 | JENNIFER NGUYEN               |

|     |                              |
|-----|------------------------------|
| 125 | JESSE THOMAS HOLZER          |
| 126 | JIAJIE CHEN                  |
| 127 | JIALE XU                     |
| 128 | JIAN WU                      |
| 129 | JIANCHANG HU                 |
| 130 | JIE SONG                     |
| 131 | JIE WEI                      |
| 132 | JIE ZHANG                    |
| 133 | JILI WANG                    |
| 134 | JINGCI MENG                  |
| 135 | JINGJIANG PENG               |
| 136 | JINGLAN LI                   |
| 137 | JIWEI ZHAO                   |
| 138 | JOHN DAVIS                   |
| 139 | JOHN GILLETT                 |
| 140 | JOHN KANE                    |
| 141 | JOHN WILTSHIRE-GORDON        |
| 142 | JONATHON PETERSON            |
| 143 | JOSEP GINEBRA                |
| 144 | JOSEPH DEUTSCH               |
| 145 | JU HEE CHO                   |
| 146 | JUN LI                       |
| 147 | JUN SHAO                     |
| 148 | JUN YIN                      |
| 149 | JUN ZHANG                    |
| 150 | JUN ZHU                      |
| 151 | JUNGWON MUN                  |
| 152 | JUNHEE HAN                   |
| 153 | JUNHO LEE                    |
| 154 | KAILEI CHEN                  |
| 155 | KAM-WAH TSUI                 |
| 156 | KARL BROMAN                  |
| 157 | KARL ROHE                    |
| 158 | KATHERINE GOODE              |
| 159 | KAZUHIKO SHINKI              |
| 160 | KEEGAN KORTHAUER             |
| 161 | KEN ONO                      |
| 162 | KEVIN HASEGAWA ENG           |
| 163 | KEVIN PACKARD                |
| 164 | KJELL DOKSUM                 |
| 165 | KRISHNAKUMAR BALASUBRAMANIAN |
| 166 | KRISTEN CYFFKA               |
| 167 | KUNLING HUANG                |

|     |                      |
|-----|----------------------|
| 168 | KYLE HEBERT          |
| 169 | KYLE HERBET          |
| 170 | KYUNGMAHN KIM        |
| 171 | LAM HO               |
| 172 | LAN LUO              |
| 173 | LANCINE KONATE       |
| 174 | LEI XU               |
| 175 | LEV BORISOV          |
| 176 | LIAM JOHNSTON        |
| 177 | LIE XIONG            |
| 178 | LILI LAN             |
| 179 | LILI ZHENG           |
| 180 | LILUN DU             |
| 181 | LIN QI               |
| 182 | LONG PHAN            |
| 183 | LU MAO               |
| 184 | LU YANG              |
| 185 | LUWAN ZHANG          |
| 186 | LUXI CAO             |
| 187 | MANJUSHA KANCHARLA   |
| 188 | MARI PALTA           |
| 189 | MARIA KAMENETSKY     |
| 190 | MARIA ROJO           |
| 191 | MARIAN R FISHER      |
| 192 | MARY LINDSTROM       |
| 193 | MATTHEW BALLARD      |
| 194 | MENG SONG            |
| 195 | MICHAEL FERRIS       |
| 196 | MICHAEL GEORGE ILLIS |
| 197 | MICHAEL HOGAN        |
| 198 | MICHAEL KUTZLER      |
| 199 | MICHAEL LIOU         |
| 200 | MICHAEL NEWTON       |
| 201 | MICHAEL RENE KOSOROK |
| 202 | MICHELLE HARRIS      |
| 203 | MIHAELA IFRIM        |
| 204 | MIN JUNG LEE         |
| 205 | MIN NIU              |
| 206 | MING XIE             |
| 207 | MING YUAN            |
| 208 | MINJING TAO          |
| 209 | MITCHELL PAUKNER     |
| 210 | MOO K CHUNG          |

|     |                         |
|-----|-------------------------|
| 211 | MUHONG GAO              |
| 212 | MURRAY CLAYTON          |
| 213 | NATALIA DE LEON GATTI   |
| 214 | NELLIE LAUGHLIN         |
| 215 | NICHOLAS HENDERSON      |
| 216 | NICHOLAS STEPHEN KEULER |
| 217 | NING FAN                |
| 218 | NORBERT BINKIEWICZ      |
| 219 | NORMAN DRAPER           |
| 220 | PAUL M TERWILLIGER      |
| 221 | PAUL RATHOUZ            |
| 222 | PAUL SAVARIAPPAN        |
| 223 | PEIGEN ZHOU             |
| 224 | PERLA REYES             |
| 225 | PHILIP WOOD             |
| 226 | QI JIANG                |
| 227 | QI TANG                 |
| 228 | QIAN ZHIGUANG           |
| 229 | QING LI                 |
| 230 | QIONG ZHANG             |
| 231 | QIURONG CUI             |
| 232 | QUEFENG LI              |
| 233 | QUOC TRAN               |
| 234 | REBECCA KOSCIK          |
| 235 | REBECCA POST            |
| 236 | RICHARD A BRUALDI       |
| 237 | RICHARD A. JOHNSON      |
| 238 | RICHARD J. CHAPPELL     |
| 239 | ROBERT HARRON           |
| 240 | ROBERT R MEYER          |
| 241 | ROBERT WARDROP          |
| 242 | ROBERT WAYNE GREEN      |
| 243 | RONALD GANGNON          |
| 244 | RONGJUN ZHU             |
| 245 | RUI CHEN                |
| 246 | RUI TANG                |
| 247 | RUIFANG SONG            |
| 248 | RUIFENG XU              |
| 249 | RUNGANG HAN             |
| 250 | RUOSI GUO               |
| 251 | RYAN ZEA                |
| 252 | SAMUEL STECHMANN        |
| 253 | SANGBUM CHOI            |

|     |                              |
|-----|------------------------------|
| 254 | SCOTT HOTTOVY                |
| 255 | SCOTT JOSEPH HETZEL          |
| 256 | SEAN KENT                    |
| 257 | SEBASTIEN ROCH               |
| 258 | SEHO PARK                    |
| 259 | SEULKEE YUN                  |
| 260 | SEUNGBONG HAN                |
| 261 | SHAN LU                      |
| 262 | SHANE HUBLER                 |
| 263 | SHANG WU                     |
| 264 | SHENG WANG                   |
| 265 | SHENG ZHANG                  |
| 266 | SHENGJI JIA                  |
| 267 | SHI JIN                      |
| 268 | SHIXUE LIU                   |
| 269 | SHIZHEN WANG                 |
| 270 | SHUAI CHEN                   |
| 271 | SHUANG HUANG                 |
| 272 | SHULEI WANG                  |
| 273 | SHUYUN YE                    |
| 274 | SIJIAN WANG                  |
| 275 | SIJING LI                    |
| 276 | SOHEIL SADEGHI               |
| 277 | SOKOL VAKO                   |
| 278 | SONA Z SWANSON               |
| 279 | SONG WANG                    |
| 280 | SRIKANTHMADHAVAN ARAVAMUTHAN |
| 281 | STEFFEN LEMPP                |
| 282 | STEPHEN AARON STANHOPE       |
| 283 | STEPHEN BERG                 |
| 284 | STEPHEN WAINGER              |
| 285 | STEPHEN WRIGHT               |
| 286 | STEVEN SAM                   |
| 287 | SUKHENDU MEHROTRA            |
| 288 | SUNDUZ KELES                 |
| 289 | TAERI UHM                    |
| 290 | TAO YU                       |
| 291 | THEVAASIINEN CHANDERENG      |
| 292 | THOMAS COOK                  |
| 293 | THOMAS G. KURTZ              |
| 294 | THU LE                       |
| 295 | TIEN VO                      |
| 296 | TIMO SEPPALAINEN             |

|     |                    |
|-----|--------------------|
| 297 | TIMOTHY IDOWU      |
| 298 | TING YE            |
| 299 | TING-LI LIN        |
| 300 | TONG LI            |
| 301 | TONGHAI YANG       |
| 302 | TRAM TA            |
| 303 | TUN LEE NG         |
| 304 | TZU HSIANG HUNG    |
| 305 | VARSHA KULKARNI    |
| 306 | VICTOR LUO         |
| 307 | VICTORIA MANSFIELD |
| 308 | WAI TONG FAN       |
| 309 | WEI ZHENG          |
| 310 | WEI-YIN LOH        |
| 311 | WENWEN ZHANG       |
| 312 | WENZHI CAO         |
| 313 | WESLEY CHANG       |
| 314 | XIAO GUO           |
| 315 | XIAO NIE           |
| 316 | XIAODAN WEI        |
| 317 | XIAOMAO LI         |
| 318 | XIAOPING FENG      |
| 319 | XIAOWU DAI         |
| 320 | XIN LI             |
| 321 | XIN ZHANG          |
| 322 | XINJIE HE          |
| 323 | XINWEI DENG        |
| 324 | XINXIN YU          |
| 325 | XINYU SONG         |
| 326 | XIPEI YANG         |
| 327 | XIRAN WANG         |
| 328 | XIUFENG SHAO       |
| 329 | XIUYU MA           |
| 330 | XIWEN MA           |
| 331 | XU HE              |
| 332 | XU XU              |
| 333 | XUEYAO CHEN        |
| 334 | XUN ZHAO           |
| 335 | YAJUAN SI          |
| 336 | YALI WANG          |
| 337 | YAN CHEN           |
| 338 | YANBING ZHENG      |
| 339 | YANG ZHAO          |

|     |                 |
|-----|-----------------|
| 340 | YANNAN QIU      |
| 341 | YAOGUO XIE      |
| 342 | YAOYAO XU       |
| 343 | YAZHEN WANG     |
| 344 | YI CHAI         |
| 345 | YI LI           |
| 346 | YI LIU          |
| 347 | YIFAN MEI       |
| 348 | YILIN ZHANG     |
| 349 | YING ZHANG      |
| 350 | YONGFENG WU     |
| 351 | YONGJOON KIM    |
| 352 | YONGSU LEE      |
| 353 | YOUNG LEE       |
| 354 | YOUNG MIN PARK  |
| 355 | YOUNGDEOK HWANG |
| 356 | YOURAN QI       |
| 357 | YU MENGANG      |
| 358 | YU QIU LIU      |
| 359 | YUAN JIANG      |
| 360 | YUAN LI         |
| 361 | YUAN WANG       |
| 362 | YUANZHI LI      |
| 363 | YUCHANG WU      |
| 364 | YUCHEN ZHOU     |
| 365 | YUJIN CHUNG     |
| 366 | YUNONG LIN      |
| 367 | YUQING XU       |
| 368 | ZHANG CHUNMING  |
| 369 | ZHENGJUN ZHANG  |
| 370 | ZHENGXIAO WU    |
| 371 | ZHIGENG GENG    |
| 372 | ZHIGUO XIAO     |
| 373 | ZHONGJIE YU     |
| 374 | ZHUANG WU       |
| 375 | ZIFENG ZHAO     |
| 376 | ZIJIAN NI       |
| 377 | ZUOFENG SHANG   |

```
quantidade <- "
  SELECT COUNT(DISTINCT instructors.id) AS QUANTIDADE FROM subjects
  INNER JOIN subject_memberships
```

```

        ON subjects.code = subject_memberships.subject_code
    INNER JOIN course_offerings
        ON subject_memberships.course_offering_uuid = course_offerings.uuid
    INNER JOIN sections
        ON course_offerings.uuid = sections.course_offering_uuid
    INNER JOIN teachings
        ON sections.uuid = teachings.section_uuid
    INNER JOIN instructors
        ON teachings.instructor_id = instructors.id
    WHERE subjects.abbreviation = 'STAT'
"

dbGetQuery(conn, quantidade)

```

#### QUANTIDADE

1            379

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?
- Quem é o professor mais fácil?
- Qual é a disciplina mais difícil?
- Qual é a disciplina mais fácil?

```

gpa <- "
SELECT
    instructors.name AS PROFESSOR,
    courses.name AS DISCIPLINA,

    (
        grade_distributions.a_count * 4 +
        grade_distributions.ab_count * 3.5 +
        grade_distributions.b_count * 3 +
        grade_distributions.bc_count * 2.5 +
        grade_distributions.c_count * 2 +
        grade_distributions.d_count * 1 +
        grade_distributions.f_count * 0
    ) * 1.0
/

```

```

        (
            grade_distributions.a_count +
            grade_distributions.ab_count +
            grade_distributions.b_count +
            grade_distributions.bc_count +
            grade_distributions.c_count +
            grade_distributions.d_count +
            grade_distributions.f_count
        ) AS GPA

FROM subjects

INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid
INNER JOIN courses
    ON course_offerings.course_uuid = courses.uuid
INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN grade_distributions
    ON sections.course_offering_uuid =
        grade_distributions.course_offering_uuid
    AND sections.number =
        grade_distributions.section_number
INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
"

dados_gpa <- dbGetQuery(conn, gpa)

professores_gpa <- "
SELECT
    PROFESSOR,
    AVG(GPA) AS GPA_MEDIO FROM (
SELECT
    instructors.name AS PROFESSOR,
    (

```

```

        grade_distributions.a_count * 4 +
        grade_distributions.ab_count * 3.5 +
        grade_distributions.b_count * 3 +
        grade_distributions.bc_count * 2.5 +
        grade_distributions.c_count * 2 +
        grade_distributions.d_count * 1 +
        grade_distributions.f_count * 0
    ) * 1.0
    /
    (
        grade_distributions.a_count +
        grade_distributions.ab_count +
        grade_distributions.b_count +
        grade_distributions.bc_count +
        grade_distributions.c_count +
        grade_distributions.d_count +
        grade_distributions.f_count
    ) AS GPA

FROM subjects

INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid =
        course_offerings.uuid
INNER JOIN courses
    ON course_offerings.course_uuid = courses.uuid
INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN grade_distributions
    ON sections.course_offering_uuid =
        grade_distributions.course_offering_uuid
    AND sections.number =
        grade_distributions.section_number
INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
)
GROUP BY PROFESSOR

```

```

ORDER BY GPA_MEDIO
"

professores_gpa_resultado <- dbGetQuery(conn, professores_gpa)

professor_dificil <- "
SELECT
    PROFESSOR,
    GPA_MEDIO
FROM (
    SELECT
        instructors.name AS PROFESSOR,

        AVG(
            (
                grade_distributions.a_count * 4 +
                grade_distributions.ab_count * 3.5 +
                grade_distributions.b_count * 3 +
                grade_distributions.bc_count * 2.5 +
                grade_distributions.c_count * 2 +
                grade_distributions.d_count * 1 +
                grade_distributions.f_count * 0
            ) * 1.0
        /
        (
            grade_distributions.a_count +
            grade_distributions.ab_count +
            grade_distributions.b_count +
            grade_distributions.bc_count +
            grade_distributions.c_count +
            grade_distributions.d_count +
            grade_distributions.f_count
        )
    ) AS GPA_MEDIO

FROM subjects

INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid =
        course_offerings.uuid

```

```

        INNER JOIN sections
            ON course_offerings.uuid = sections.course_offering_uuid
        INNER JOIN grade_distributions
            ON sections.course_offering_uuid =
                grade_distributions.course_offering_uuid
            AND sections.number =
                grade_distributions.section_number
        INNER JOIN teachings
            ON sections.uuid = teachings.section_uuid
        INNER JOIN instructors
            ON teachings.instructor_id = instructors.id
        WHERE subjects.abbreviation = 'STAT'
        GROUP BY instructors.id
    )
    ORDER BY GPA_MEDIO
    LIMIT 1
"

dbGetQuery(conn, professor_dificil)

```

```

        PROFESSOR GPA_MEDIO
1 CHRISTINE SORKNESS      NA

```

```

professor_facil <- "
    SELECT
        PROFESSOR,
        GPA_MEDIO
    FROM (
        SELECT
            instructors.name AS PROFESSOR,

            AVG(
                (
                    grade_distributions.a_count * 4 +
                    grade_distributions.ab_count * 3.5 +
                    grade_distributions.b_count * 3 +
                    grade_distributions.bc_count * 2.5 +
                    grade_distributions.c_count * 2 +
                    grade_distributions.d_count * 1 +
                    grade_distributions.f_count * 0
                ) * 1.0
            )

```

```

        /
        (
            grade_distributions.a_count +
            grade_distributions.ab_count +
            grade_distributions.b_count +
            grade_distributions.bc_count +
            grade_distributions.c_count +
            grade_distributions.d_count +
            grade_distributions.f_count
        )
    ) AS GPA_MEDIO

FROM subjects

INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid =
        course_offerings.uuid
INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN grade_distributions
    ON sections.course_offering_uuid =
        grade_distributions.course_offering_uuid
    AND sections.number =
        grade_distributions.section_number
INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.id
)
ORDER BY GPA_MEDIO DESC
LIMIT 1
"

dbGetQuery(conn, professor_facil)

```

PROFESSOR GPA\_MEDIO  
1 SRIKANTHMADHAVAN ARAVAMUTHAN 4

```

disciplinas_gpa <- "
SELECT
  courses.name AS DISCIPLINA,
  AVG(
    (
      grade_distributions.a_count * 4 +
      grade_distributions.ab_count * 3.5 +
      grade_distributions.b_count * 3 +
      grade_distributions.bc_count * 2.5 +
      grade_distributions.c_count * 2 +
      grade_distributions.d_count * 1 +
      grade_distributions.f_count * 0
    ) * 1.0
    /
    (
      grade_distributions.a_count +
      grade_distributions.ab_count +
      grade_distributions.b_count +
      grade_distributions.bc_count +
      grade_distributions.c_count +
      grade_distributions.d_count +
      grade_distributions.f_count
    )
  ) AS GPA_MEDIO

FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid =
    course_offerings.uuid
INNER JOIN courses
  ON course_offerings.course_uuid = courses.uuid
INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN grade_distributions
  ON sections.course_offering_uuid =
    grade_distributions.course_offering_uuid
  AND sections.number =
    grade_distributions.section_number
WHERE subjects.abbreviation = 'STAT'

```

```

GROUP BY courses.uuid
ORDER BY GPA_MEDIO
"

dbGetQuery(conn, disciplinas_gpa)

```

|    | DISCIPLINA                                                       | GPA_MEDIO |
|----|------------------------------------------------------------------|-----------|
| 1  | Directed Study                                                   | NA        |
| 2  | Practicum in Clinical Trial Data Analysis and Interpretation     | NA        |
| 3  | Research                                                         | NA        |
| 4  | Statistical Methods for Medical Image Analysis                   | NA        |
| 5  | Directed Study                                                   | NA        |
| 6  | Senior Honors Thesis                                             | NA        |
| 7  | Senior Honors Thesis                                             | NA        |
| 8  | Introduction to the Theory of Probability                        | 2.891568  |
| 9  | Linear Programming Methods                                       | 2.959583  |
| 10 | Introduction to Probability and Mathematical Statistics I        | 3.031950  |
| 11 | Introductory Statistics for Engineers                            | 3.039691  |
| 12 | Introduction to Stochastic Processes                             | 3.059488  |
| 13 | Introduction to Statistical Methods                              | 3.076261  |
| 14 | Introduction to Probability and Mathematical Statistics II       | 3.097313  |
| 15 | Introduction to Combinatorics                                    | 3.117675  |
| 16 | Introduction to Theory and Methods of Mathematical Statistics I  | 3.154021  |
| 17 | Introductory Applied Statistics for Engineers                    | 3.178822  |
| 18 | An Introduction to Sample Survey Theory and Methods              | 3.210439  |
| 19 | Introductory Applied Statistics for the Life Sciences            | 3.223803  |
| 20 | Accelerated Introduction to Statistical Methods                  | 3.250943  |
| 21 | Applied Regression Analysis                                      | 3.308202  |
| 22 | Applied Categorical Data Analysis                                | 3.327047  |
| 23 | Introduction to Theory and Methods of Mathematical Statistics II | 3.363738  |
| 24 | Introduction to Time Series                                      | 3.396354  |
| 25 | Statistical Experimental Design                                  | 3.419259  |
| 26 | Theory of Probability I                                          | 3.425017  |
| 27 | Intro to Biostats for Pharmacy                                   | 3.426641  |
| 28 | Financial Statistics                                             | 3.436086  |
| 29 | Applied Multivariate Analysis                                    | 3.437232  |
| 30 | Nonlinear Optimization I                                         | 3.447456  |
| 31 | Introduction to Computational Statistics                         | 3.458916  |
| 32 | Statistical Methods for Bioscience I                             | 3.489562  |
| 33 | Mathematical Statistics I                                        | 3.502526  |
| 34 | Statistical Methods for Bioscience II                            | 3.513428  |
| 35 | Statistical Methods II                                           | 3.520567  |

|    |                                                                  |          |
|----|------------------------------------------------------------------|----------|
| 36 | Theory of Probability II                                         | 3.538027 |
| 37 | Mathematical Statistics                                          | 3.556040 |
| 38 | Statistical Methods I                                            | 3.564059 |
| 39 | Theory and Application of Regression and Analysis of Variance II | 3.595450 |
| 40 | Introduction to Statistical Inference                            | 3.596234 |
| 41 | Mathematical Statistics                                          | 3.600373 |
| 42 | Learning a Statistical Language                                  | 3.600436 |
| 43 | Theory and Application of Regression and Analysis of Variance I  | 3.611908 |
| 44 | Statistical Methods for Clinical Trials                          | 3.613813 |
| 45 | Introduction to Clinical Trials I                                | 3.624917 |
| 46 | Professional Skills in Data Science                              | 3.637931 |
| 47 | Theory of Probability                                            | 3.646316 |
| 48 | Introduction to Bayesian Decision and Control I                  | 3.666382 |
| 49 | Generalized Linear Models                                        | 3.680000 |
| 50 | Special Topics in Statistics                                     | 3.684694 |
| 51 | Statistical Consulting                                           | 3.695769 |
| 52 | Introduction to Biostatistics                                    | 3.713258 |
| 53 | Statistical Methods for Epidemiology                             | 3.748634 |
| 54 | Applied Time Series Analysis, Forecasting and Control I          | 3.753848 |
| 55 | Statistical Methods for Spatial Data                             | 3.753965 |
| 56 | Statistical Computing                                            | 3.755939 |
| 57 | Introduction to Biostatistical Methods for Public Health         | 3.769922 |
| 58 | Special Topics in Statistics                                     | 3.798822 |
| 59 | Theory of Probability                                            | 3.810367 |
| 60 | Special Topics in Statistics                                     | 3.820894 |
| 61 | Multivariate Analysis I                                          | 3.830763 |
| 62 | Large Sample Theory of Statistical Inference                     | 3.830808 |
| 63 | Survival Analysis Theory and Methods                             | 3.861259 |
| 64 | Statistical Methods for Molecular Biology                        | 3.869444 |
| 65 | Introductory Nonparametric Statistics                            | 3.884375 |
| 66 | Nonlinear Regressn Analysis                                      | 3.909091 |
| 67 | Topics in the Theory of Probability                              | 3.931512 |
| 68 | Seminar                                                          | 3.957715 |
| 69 | Statistical Model Building and Learning                          | 3.967902 |
| 70 | Decision Trees for Multivariate Analysis                         | 3.968244 |
| 71 | Non Parametric Statistics                                        | 3.971397 |
| 72 | Estimation of Functions from Data                                | 3.977381 |
| 73 | Experimental Design I                                            | 3.982051 |
| 74 | Nonparametric Statistics and Machine Learning Methods            | 4.000000 |
| 75 | Sample Survey Theory and Method                                  | 4.000000 |
| 76 | Data Science Practicum                                           | 4.000000 |
| 77 | Empir Proc&Semiparmtrc Infernc                                   | 4.000000 |

```

disciplina_facil <- "
SELECT
    courses.name AS DISCIPLINA,

    AVG(
        (
            grade_distributions.a_count * 4 +
            grade_distributions.ab_count * 3.5 +
            grade_distributions.b_count * 3 +
            grade_distributions.bc_count * 2.5 +
            grade_distributions.c_count * 2 +
            grade_distributions.d_count * 1 +
            grade_distributions.f_count * 0
        ) * 1.0
        /
        (
            grade_distributions.a_count +
            grade_distributions.ab_count +
            grade_distributions.b_count +
            grade_distributions.bc_count +
            grade_distributions.c_count +
            grade_distributions.d_count +
            grade_distributions.f_count
        )
    ) AS GPA_MEDIO

FROM subjects

INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid =
        course_offerings.uuid
INNER JOIN courses
    ON course_offerings.course_uuid = courses.uuid
INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN grade_distributions
    ON sections.course_offering_uuid =
        grade_distributions.course_offering_uuid
AND sections.number =
    grade_distributions.section_number

```

```

WHERE subjects.abbreviation = 'STAT'
GROUP BY courses.uuid
ORDER BY GPA_MEDIO DESC
LIMIT 1
"

```

```
dbGetQuery(conn, disciplina_facil)
```

```

              DISCIPLINA GPA_MEDIO
1 Empir Proc&Semiparmtrc Infernc      4

```

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-24 00:44:33 -03"
```